

Task definition

Business value

Clarity of definition

Alternatives

Data

pretraining
human annotations

Expert

MTurk

data augmentations

weak supervision study

sampling from business traffic

Model

Input/output representation

Tokenization (char, word)

FFNN, RNN, CNN

Transformers

Multi modal

Loss

Cross entropy, hinge, ...

Regularizations

- Batch norm

- Layer norm

- L1/L2

- Dropout

Multi task facilitation

Contrastive learning

Optimization

Gradient descent

SGD

Minibatch GD

Momentum, ADAM

Validation

Intrinsic: over/underfitting

Extrinsic: OOO test

Efficiency

Latency / throughput
Dollar cost

Business

Short-term

- KPIs
- Collateral damage evaluation
- External impact
- Statistical testing
- Model calibration

Long-term

- Model maintenance
 - refresh
 - failure logic
 - Continuous perf. tracking
- Custom behavior impact
- Public relation